

IOCTreaper – Application Requirements Specifications

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Introduction

This Specification requirement document defines and documents the functional requirements for a web application intended to extract Indicators of Compromise from a raw text named IOCTree .[1]

1.1 Purpose

IOCTree is intended to extract common indicator types from raw text input provided by a user . This document captures a comprehensive list of functionalities of the application , maps them to defined actions to be taken and defines the successful cases and failure cases for each actions . This document is intended to provide design and developmental clarity to the application development teams , serve as a reference guide to architectural planning and development stakeholders .

1.2 Scope

This document delves into individual application functionality of IOCTree , while defining conditions for a successful vs unsuccessful operational executions , differences in flows of executing the same tasks and finally lists out error cases for each functionalities .

1.3 Definitions, acronyms, and abbreviations

- *Abbreviations*

- IOC – Indicator of Compromise
- IOA - Indicator of Attack
- TI – Threat Intelligence
- Webapp – Web Application
- IPv4 – Internet Protocol Address Version 4.0
- URL – Uniform Resource Locator

- *Definitions*

- Domain – A Domain is defined as a string of text that is mapped to an IP Address .[2]
- URL – A URL is a web address that provides a specific location for a particular resource on the internet . It also contains further details like the protocol to be used to access the resource etc . [3]
- Hashing - The process of converting data into an alphanumeric string of fixed length using a mathematical algorithm in an irreversible process is called Hashing . [4]

- *Acronyms*

- Use Case 1 – Use case descriptor for the action of upload and extract IOCs from raw text . [5]
- Use Case 2 - Use case descriptor for the action to export extracted IOCs.
- Use Case 3 - Use case descriptor for the action to search for a specific IOC in an uploaded text .
- Use Case 4 - Use case descriptor for the action to tag IOCs by Type .
- Use Case 5 – Use Case descriptor for copying text to clipboard rather than exporting data into a predefined format .
- Use Case 6 – Use Case descriptor for deleting text from the input text window .
- Use Case 7 – Use Case descriptor for describing errors during application run time .
- Use Case 8 - Use Case descriptor for describing exporting of tags along with IOCs .

2. Overview

This application is meant to serve as an investigation enabler to security and threat intelligence analysts by aiding them in quickly parsing and extracting information of use from different text formats . This application aims to achieve this goal broadly as follows:

1. Taking raw data input of fixed length from an analyst .
2. Parsing this information through predefined conditions and patterns
3. Identifying and categorizing IOCs based on their defined type
4. Displaying this segregated information for providing data clarity to the user .
5. Allows for searching and tagging of data to provide more context around an IOC for the analyst .
6. Supports copying and exporting of information for further data transformation or utilization by the analyst .

2.1 Product perspective

IOCREAPER is meant to act as a standalone application which can be utilized for several requirement scenarios . This tool is meant to be deployed on individual systems which can be initialized on demand with the individual users controlling and executing actions

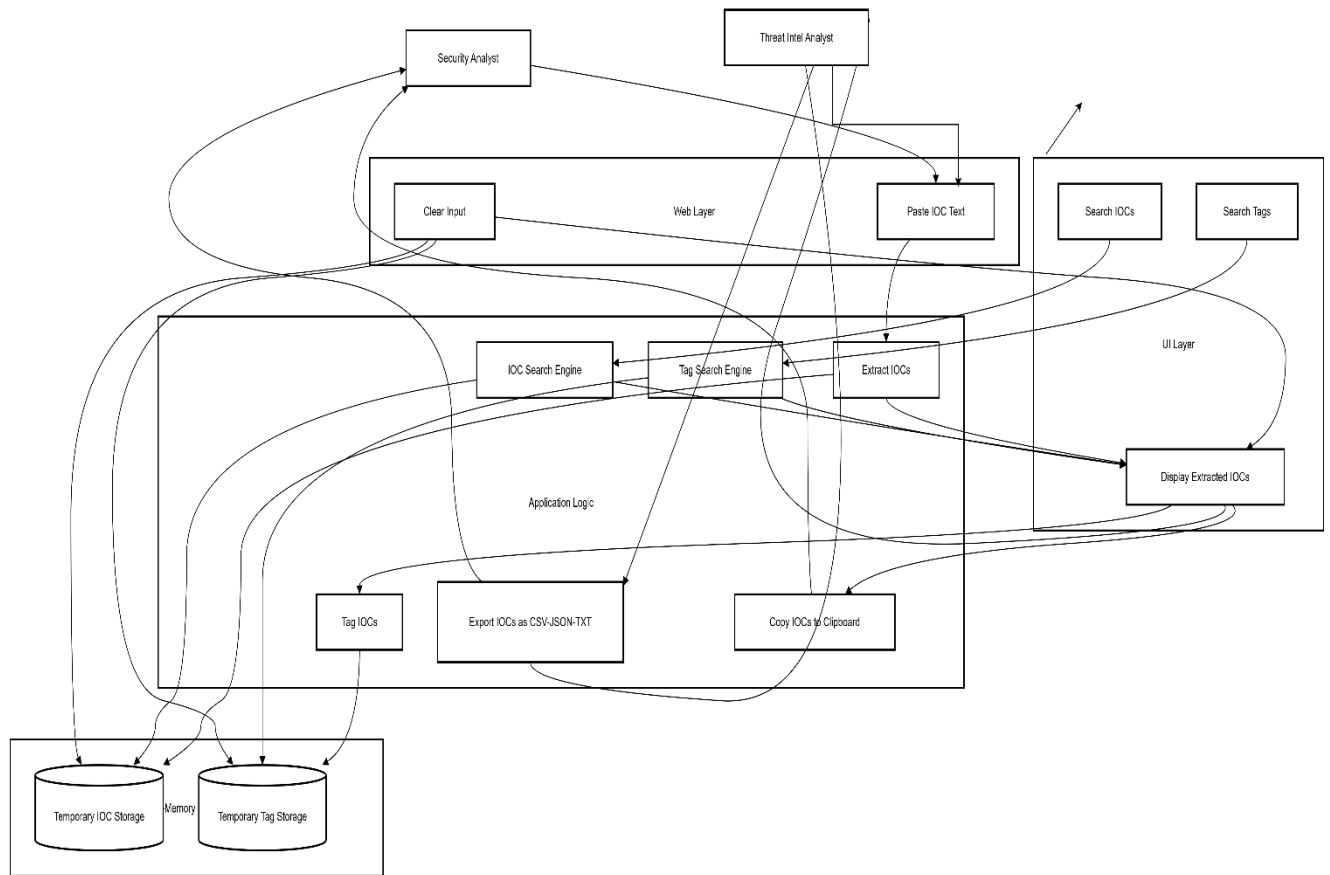


Figure 1. Product Interlinking perspective

2.2 Product functions

2.2.1 Requirement ID: FR-001

FR-001: Parse IOCs from Raw Text

Use Case Reference: Use Case 1.

Trigger: User Copies and Pastes Text into an input area and clicks “Extract”.

Preconditions:

1. Application must be in Active or Run State on the host .
2. The text size must be within the acceptable limits of ~ 100 KB Length
3. System has enough memory space to run the application .

Success Flow:

1. The application first validates the input length / size .
2. User should correctly click on the Extract option .
3. The application run regex-based extraction .

4. If selected the data will be refanged or normalized to their usual nature .
5. Also, if selected the output values will be de-duplicated to get only unique values .
6. The extracted data is grouped according to the types of IOCs and shown along with their values .

Alternative Flows:

1. Instead of pressing “Extract” option , user might press Enter or Ctrl+ Enter to trigger the parsing action .
2. Instead of clicking with the mouse pointer , user might use Tab key for button navigation .

Postconditions:

1. IOCs are available for next set of data transformation or other supported actions

Error Flows:

1. Empty data input → User is asked to enter the text for further operations to be performed .
2. Oversized Input → User is asked to resupply the text input to match the application requirement .
3. Application Timeout → User is requested to initiate the extraction process again .

2.2.2 Requirement ID: FR-002

FR-002: Categorize Extracted IOCs

Use Case Reference: Use Case 2

Trigger: The application completes the data extraction and parsing successfully .

Preconditions:

1. At least one supported IOC category is identified.

Success Flow:

1. System categorized and calculated unique value of each IOC (IPv4, IPv6, Domain, URL, Email, MD5/SHA1/SHA256).
2. These values are displayed in easily identifiable and navigable format with their counts.

Alternative Flows:

1. If no IOC is identified, then the user is notified about the lack of IOCs in the text .

Postconditions:

1. User is shown the structured results with options for facilitating further data transformation or exporting .

Error Flows:

1. Data text with no supported IOC text is supplied → User is notified that there were no identifiable IOCs observed .

2.2.3 Requirement ID: FR-003

FR-003: Apply Filters and Normalization

Use Case Reference: Use Case 2

Trigger:

1. The application completes the data extraction and parsing successfully

Preconditions:

1. Input text provided.
2. At least one supported IOC category is identified.

Success Flow:

1. Upon entering the text into the input text field, the user is provided with an option to modify the text specifically to complete one of the actions from below
2. Filters or transformation operations: de-duplication, refang, normalize or defang.
3. The data is parsed with these options and.
4. The user is provided with an option to export the text for further uses .

Alternative Flows:

1. The user chooses to just parse the text with no options applied.

Postconditions:

1. The user is shown results reflecting their chosen filters or data transformation .

Error Flows:

1. The user chooses an invalid option combinations → system notifies the user that the operations cannot be performed in the chosen sequence.

2.2.4 Requirement ID: FR-004

FR-004: Search Results

Use Case Reference: Use Case 3

Trigger: User types into search box.

Preconditions: IOC results already displayed.

Success Flow:

1. The application accepts user query string.
2. If a matching result is observed, then the results are highlighting .

Alternative Flows:

1. User would need to select case-sensitive text option for finding exact matches
2. Similar matches are found in different IOCs categories all of which should be displayed .

Postconditions:

1. Matching results will be highlighted for further actions .

Error Flows:

1. If No matches are found → user is notified with a “No results found” message .

2.2.5 Requirement ID: FR-005

FR-005: Copy to Clipboard

Use Case Reference: Use Case 5

Trigger:

1. The user clicks “Copy IOCs” or selects “Copy” action for specific category of IOCs .

Preconditions:

1. Supported IOC patterns must be identified in the input text.

Success Flow:

1. Once the “Copy” action is selected the selected IOC are copied.
2. This copy action will be in a newline-separated values to clipboard.
3. A confirmation message is shown to the user .

Alternative Flows:

1. User uses keyboard shortcut for copying action Ctrl + C.

Postconditions:

1. The selected IOCs stored in clipboard for further action by the user .

Error Flows:

1. If the clipboard API fails → instructions to copy manually are shown .
2. User might use Cut action instead of copy action → User is notified that they have initiated an unsupported action .

2.2.6 Requirement ID: FR-006

FR-006: Export Results to File

Use Case Reference: Use Case 1

Trigger:

1. The user selects “Export” option and chooses format (CSV, JSON, TXT).

Preconditions:

1. There should be at least one IOC present.

Success Flow:

1. Upon selection of “Export” action the application will serialize IOC list into chosen format.
2. The list will then be provided in the specified format for download to user.

Alternative Flows:

1. The user exports specific category only.
2. The user copies the IOCs into clipboard directly .

Postconditions:

1. The file downloaded successfully.

Error Flows:

1. If No IOCs are available, → The Export option is disabled or error shown.

2.2.7 Requirement ID: FR-007

FR-007: Clear Session

Use Case Reference: Use Case 6

Trigger:

1. Upon completion of the parsing actions the user clicks “Clear” button.

Preconditions:

1. There should be at least one IOC result.

Success Flow:

1. In memory IOC storage is cleared .
2. The UI is reverted to its initial state.

Alternative Flows: None.

Postconditions:

1. There should no results or inputs remaining ; the input should be ready for new data input.

Error Flows: None.

2.2.8 Requirement ID: FR-008

FR-008: Session-Scoped Data Handling

Use Case Reference: Use Case 1

Trigger:

1. The session begins when user opens app.

Preconditions:

1. The application should be installed on the user`s machine .
2. There should be at least one browser to open the application .

Success Flow:

1. The IOCs are stored within memory for one session only.
2. The data is reset when session ends, or user clears results.

Alternative Flows: None.

Postconditions:

1. There should be no persistent storage of user data.

Error Flows: None.

2.2.9 Requirement ID: FR-009

FR-009: Input Validation & Size Limits

Use Case Reference: Use Case 1

Trigger:

1. The user submits an input text.

Preconditions:

1. The Input text should be provided.

Success Flow:

1. The application will validate the size of the text to be around ~ 100 KB or not .
2. If the input is oversized , then it is Rejected with a custom message.

Alternative Flows: None .

Postconditions:

1. The data is parsed and made ready for exporting or copying by the user .

Error Flows:

1. If the input is oversized it is rejected.

2.2.10 Requirement ID: FR-010

FR-010: Error Messaging

Use Case Reference: Use Case 7

Trigger:

1. Any type of error might occur during parsing, search, copy, tagging or exporting of data .

Preconditions:

1. A user specified action executed.

Success Flow:

1. There should be successful detection of error condition.
2. This interruption of the application run is shown in a user-friendly error message.
3. The user must be allowed retry without crashing.

Alternative Flows:

1. If the application is being debugged it must be allowed to log error for debugging in dev mode.

Postconditions:

1. The user is informed of issue.

Error Flows:

1. None beyond intended error reporting.

2.2.11 Requirement ID: FR-011

FR-011: Add Tags to IOCs

Use Case Reference: Use Case 4

Trigger:

1. The user selects one or more IOCs and applies a tag from the UI.

Preconditions:

1. IOCs must be parsed from the text input .
2. IOC results must already be displayed .

Success Flow:

1. The user selects one or more IOCs.
2. Then they enter a new tag or selects from existing tags.
3. System attaches the tag to the selected IOC(s).

Alternative Flows:

1. Multiple tags can be applied to one IOC at once.
2. The user reuses an existing tag.

Postconditions:

1. IOCs must show associated tags in results view.

Error Flows:

1. User provides invalid input (ex:- empty tag, unauthorized special chars) → An error message is shown to the user that the tag has not been applied to the IOC.

2.2.12 Requirement ID: FR-012

FR-012: Manage Tags

Use Case Reference: Use Case 4

Trigger:

1. The user opens the Tag Management option / section in the application.

Preconditions:

1. There should be at least one tag applied to an IOC.

Success Flow:

1. The application displays a list of all tags used in the session.
2. It can allow the user to rename or delete tags.

3. If there is an modify or delete action, the application will update all the IOCs associated with modified or removed tags.

Alternative Flows: None.

Postconditions:

1. A special list should reflect all user modifications.

Error Flows:

1. The user attempts to delete a tag in use → User is requested to confirm with a notification .
2. The user attempts to provide an empty tag name → User is notified with an error message.
3. The user attempts to provide a tag name with forbidden characters → User is notified with an error message.

2.2.13 Requirement ID: FR-013

FR-013: Export Tags with IOCs

Use Case Reference: Use Case 8

Trigger:

1. The user should click Export after tagging IOCs.

Preconditions:

1. There should be at least one IOC with a tag.

Success Flow:

1. All the tags are included as an additional column or field in exported file (CSV/JSON/TXT).
2. Validation is put in place to ensure tags align with correct IOC values.

Alternative Flows:

1. The user selects “Export without tags” option .
2. The user selects “Copy to Clipboard” option .

Postconditions:

1. The exported file includes IOC along with their tag mappings.

Error Flows:

1. If no tags are present → the export action proceeds without tags.

2.3 User Characteristics

We have two user categories scoped to use this application .

1. Security Analysts

2. Threat Intel Analysts

Both the user categories are expected to be technically proficient meaning able to acquire , deploy and navigate through the application with minimum technical walkthrough of the application .

Also, users from both the categories would be able to understand and follow instructions or notifications from the application such as error messages to perform basic repetition of previous state of actions .

Finally, the utilization of the application in terms of use cases have been judged to be similar for both the user classes , as their job duties affect the data manipulation further downstream in terms of the data life cycle .

3. External Interface Requirements

- Web UI: A single-page dashboard accessed via a default browser on local host
 - Input: An input field with support to multi-line text area for raw text paste action.
 - Controls:
 1. Parsing text
 2. Clear the text input after completion of parsing action
 3. An Export dropdown action
 4. Copy button to copy text into clipboard
 5. A search box to search for IOCs and tags.
 6. Input text area to add tags beside IOCs .
 - Output:
 1. A tabbed/sectioned/windowed results page with unique counts of IOCs;
 2. A tabbed/sectioned/windowed results for viewing and searching for Tags or IOCs
 3. A tabbed/sectioned/windowed results for adding tags beside the IOCs
 4. Downloadable files in CSV/JSON/Text format.
 - No external APIs: Network calls are disabled/not used.
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4. References

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